JAEWON SEO

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EDUCATION

POSTECH (Pohang University of Science and Technology) *2023 – Present*
B.S. in Industrial and Management Engineering, Double Major in Mathematics

EXPERIENCE

Samsung Fire & Marine Insurance — Artificial Intelligence Engineer Intern *Jun – Aug 2026*

- Developed AI technologies for applications in the insurance domain

Nadoomodoo Inc. — Software Engineer Intern *Jan – Jun 2026*

- Built full-stack features (TypeScript, React) across Nadooio (website-builder SaaS) and ECSO (CSO-facing ERP), including an AI-provider routing layer for LLM and text-to-image model calls

Financial Big-Data Analysis Laboratory (FBDA Lab), POSTECH — Undergraduate Researcher *Jun – Jul 2025*

- Reproduced and extended a factor-and-idiosyncratic VAR (FIVAR) model for large-dimensional volatility matrix estimation on high-frequency S&P 500 data, implementing Huber-LASSO regression in Python/R to handle heavy-tailed returns

Statistics and Data Science Laboratory (SDS Lab), POSTECH — Undergraduate Researcher *Dec 2024 – Feb 2025*

- Researched diffusion model guidance methods (classifier-free guidance, autoguidance) and proposed a heavy-tailed guidance scheme combined with sigma/weight scheduling to improve sample diversity, validated on toy manifold and Stable Diffusion experiments

AWARDS

Excellence Award — The 3rd Wind Power Generation Forecasting AI Competition (BARAM 2026) *Sep 2026*
Korea East-West Power Co., Ltd.

- Forecasted hourly power output of the Taebaek Gadeoksan Wind Farm; ranked 9th among 986 teams on the final (private) leaderboard

3rd Prize — POSTECH Undergraduate Research Program (UGRP) *Feb 2025*

- LLM traffic optimization using prompt engineering and multi-modal features

3rd Prize — OIBC Big Data Challenge *Dec 2024*

- Predicted System Marginal Price (SMP) in the Jeju Island electricity market; ranked 6th among 207 teams

2nd Prize — Crackathon *Nov 2024*

- Ideathon proposing new lines of business for global companies; ranked 2nd among 6 teams

SKILLS

Machine Learning & Deep Learning: Python, PyTorch, R
Web Development: TypeScript, React, AWS

TEACHING & MENTORING

Student Mentor, Calculus I — POSTECH *Spring 2024, Spring 2025*
Student Mentor, Calculus II — POSTECH *Fall 2024, Fall 2025*

LANGUAGES

Korean *Native*
English *Professional working proficiency — TEPS 447 (Aug 2026)*